

Report of
Career Link

Submitted
To

School of Computer Science and Engineering

IILM University, Greater Noida, U.P.

In partial fulfilment of the requirements for the degree of

MCA



By

Roll Number of Student: 25scs2040003487

Name of Student: SAMIA KHAN

Batch:

2026-27

CANDIDATE'S DECLARATION

I, Samia Khan, hereby declare that the internship report titled “**Career Link**” has been completed by me in fulfilment of the requirements for the award of the degree of **Master of Computer Applications (MCA) in Computer Science and Engineering (CSE)**. All references and sources used in this report have been duly acknowledged and cited. I affirm that this report is my original work and has not been submitted to any other institute or university for the fulfilment of any degree or diploma requirement. I shall be solely responsible for any copyright violation arising in connection with this report.

Signature

Student Name _____

Roll No. _____

ACKNOWLEDGEMENT

I take this opportunity to express my sincere gratitude to everyone who supported me throughout the Internship Program. I am deeply thankful for their inspiring guidance, invaluable constructive criticism, and friendly advice during the course of this work. I am sincerely grateful to everyone who shared their valuable and insightful views on various aspects related to this work.

I would also like to express my heartfelt gratitude to my parents for their constant support, encouragement, and blessings throughout this journey.

Signature

Student Name _____

Roll No.. _____

3. Internship Completion Certificate



4. Project Description

4.1 Introduction

Career Link is a web-based job search platform designed to help students and job seekers find suitable off-campus employment opportunities. The platform provides job listings, company information, job descriptions, and interview-related insights in one place.

Career Link also includes an AI-based job recommendation feature that matches users with relevant job opportunities based on their skills and resume. The platform aims to simplify the job search process, reduce job-role mismatch, and help candidates make informed career decisions.

The system provides a user-friendly interface for job seekers to explore opportunities while enabling companies to connect with suitable candidates efficiently. Overall, Career Link aims to make the job search and recruitment process more accessible, organized, and personalized.

4.2 Organization Profile

Codec Technologies is an IT and technology training organization that provides internship programs, technical courses, and skill-development opportunities for students and beginners. The organization offers programs in areas such as Web Development, MERN Stack Development, Artificial Intelligence, Python Development, Data Analytics, Cybersecurity, and other emerging technologies.

Codec Technologies provides practical learning through structured training programs and project-based internships. Its internship offerings include opportunities for students to gain experience in frontend and backend technologies, databases, frameworks, and other industry-relevant skills. A MERN Stack Developer internship offered by the organization covers technologies including MongoDB, Express.js, React.js, Node.js, JavaScript, HTML, CSS, and Git/GitHub.

Codec Technologies is listed on the **AICTE Internship Portal** as an internship provider, with internship opportunities available in different technology domains.

The organization focuses on helping students develop practical technical skills through internships, learning resources, mentorship, and project-based work. Its internship programs are designed to provide learners with exposure to real-world technologies and help them prepare for future academic and professional opportunities.

4.3 Problem Statement

Students and fresh graduates often face difficulties in finding suitable off-campus job opportunities because job information is distributed across multiple platforms and sources. It can be challenging for candidates to identify job roles that match their skills, qualifications, and career interests. Additionally, limited access to company information and interview-related insights can make it difficult for students to make informed career decisions.

Another major challenge is that traditional job-search platforms mainly focus on displaying available vacancies, leaving candidates to manually determine whether they are suitable for a particular role. This can result in candidates applying to irrelevant positions, wasting time, and missing opportunities that actually match their skill sets. There is a need for a more personalized approach that goes beyond simply listing jobs and helps candidates understand the relevance of an opportunity to their existing skills.

Career Link addresses this gap by providing a centralized platform for job discovery along with an AI-based recommendation system that analyzes a user's resume and skills to identify relevant job opportunities. By connecting the candidate's existing skill profile with available job requirements, the system aims to make job discovery more targeted, reduce job-role mismatch, and help candidates focus their efforts on opportunities where they are more likely to be suitable.

4.4 Project Objectives

- Bridge the gap between students and companies by providing a structured platform where students can explore job opportunities and companies can discover skilled candidates.

- Enhance job search efficiency using AI-powered job recommendations that match users with suitable roles based on their resumes and skills.
- Offer significant insights through detailed company profiles, job descriptions, and interview details to help students make informed decisions.
- Facilitate seamless hiring processes by allowing companies to create profiles, track interested candidates, and streamline recruitment through AI-driven candidate matching.
- Keep users updated on opportunities via a subscription system for real-time job alerts, hiring trends, and interview insights.
- Address common job market challenges such as job-role mismatch and difficulty finding job openings.
- Ensure an easy-to-use and modern platform with a user-friendly interface for smooth navigation and accessibility.

4.5 Scope of the Project

The scope of **Career Link** covers the development of a web-based platform that simplifies the job search and recruitment process for students, job seekers, and companies. The platform provides a centralized environment for discovering job opportunities, accessing company information, and connecting suitable candidates with relevant job roles.

The major areas covered by the project include:

- **Job Search and Discovery:** Users can browse and explore available job opportunities based on different requirements.
- **AI-Based Job Recommendations:** The system can analyze user resumes and skills to recommend relevant job opportunities.
- **Company Information:** Users can access company profiles and relevant information to better understand potential employers.
- **Job and Interview Insights:** The platform provides job descriptions and interview-related information to help candidates prepare and make informed decisions.
- **User and Company Profiles:** Candidates and companies can maintain profiles containing relevant information.

- **Candidate Matching:** The system aims to assist companies in identifying candidates whose skills are relevant to their job requirements.
- **Job Alerts and Subscriptions:** Users can subscribe to receive updates about new opportunities and relevant job information.
- **User-Friendly Interface:** The platform is designed with a simple and modern interface to provide easy navigation and accessibility.

The project can be further expanded in the future by integrating real-time job APIs, advanced AI-based skill-gap analysis, automated resume evaluation, interview preparation features, application tracking, and personalized career guidance. These enhancements can make Career Link a more comprehensive career and recruitment platform.

4.6 Technologies and Tools Used

Career Link was developed using the MERN stack along with supporting libraries and tools for styling, validation, and animation:

- Frontend: Next.js (React.js) with Tailwind CSS for a responsive, modern user interface.
- Backend: Node.js with Express.js for RESTful API development.
- Database: MongoDB (NoSQL, hosted on MongoDB Atlas) for flexible, document-based data storage.
- UI/UX Enhancements: Framer Motion for animations, and Formik with Yup for form handling and validation.
- Development & Collaboration Tools: VS Code, GitHub for version control, and Postman for API testing.
- Deployment: Vercel for frontend hosting, with backend deployment options such as Render or Heroku.

4.7 System Architecture

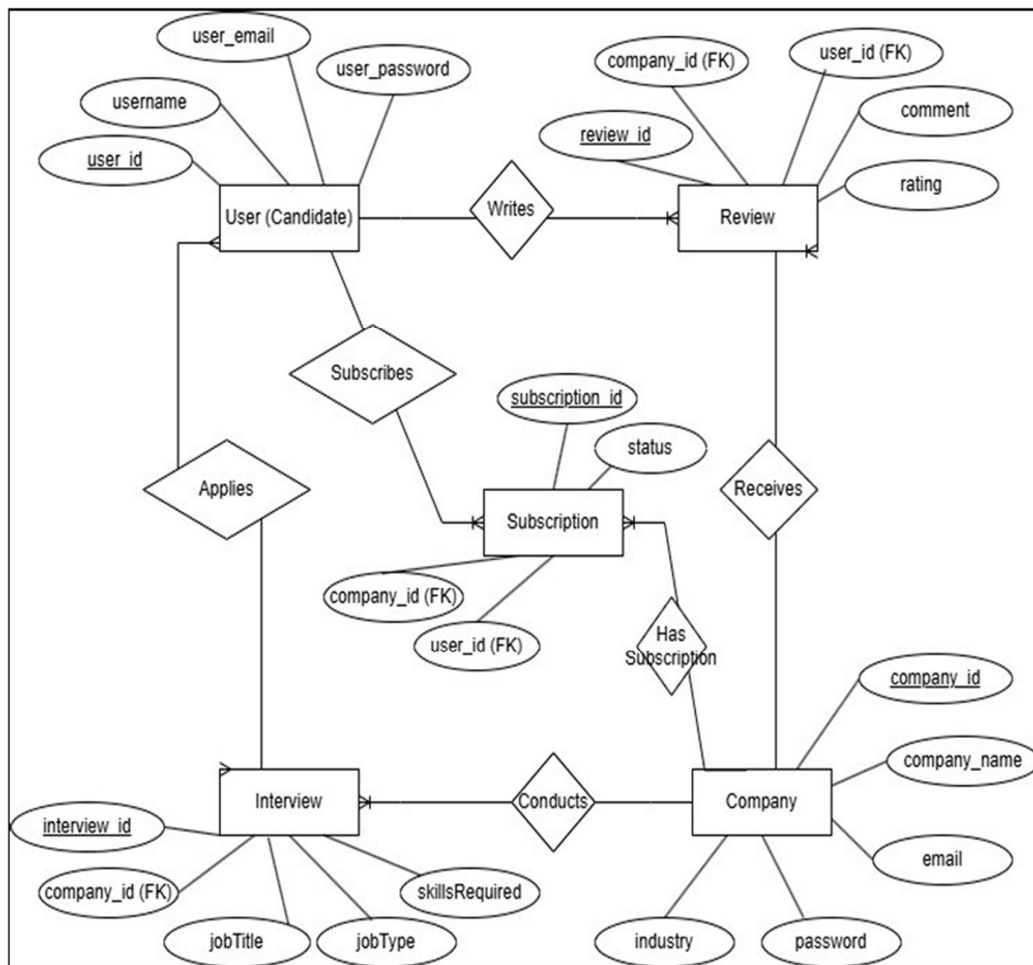
Career Link follows a client-server architecture built on the MERN stack, divided into three primary layers:

- Frontend (Client-side): Includes the Homepage, Student Dashboard, Company Dashboard, a Search & Filter module, and an Interview Details page, all built using Next.js and styled with Tailwind CSS.

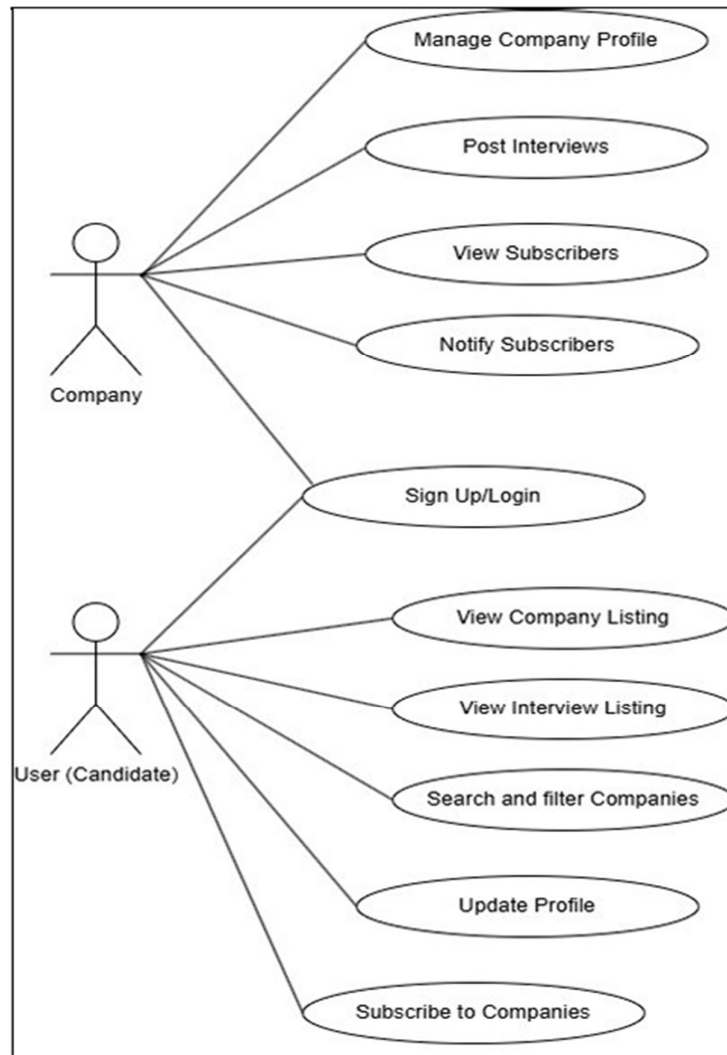
- Backend (Server-side): Comprises an Authentication Module (JWT-based login/signup for students and companies), a Company Module, a Student Module, a Subscription & Notification Module, and a Search API for filtered results.
- Database: Organized into Users, Companies, Interviews, and Subscriptions collections in MongoDB, capturing entities such as candidates, companies, interviews, reviews, and subscriptions along with their relationships.

The system design was supported by an Entity Relationship (ER) Diagram to define the database structure, a Use Case Diagram to capture the functional requirements of each actor, and multi-level Data Flow Diagrams (Zero-level, Level 1) to represent how registration, application, subscription, and review data move through the system.

- **Entity Relationship (ER) Diagram**



- **Use Case Diagram**



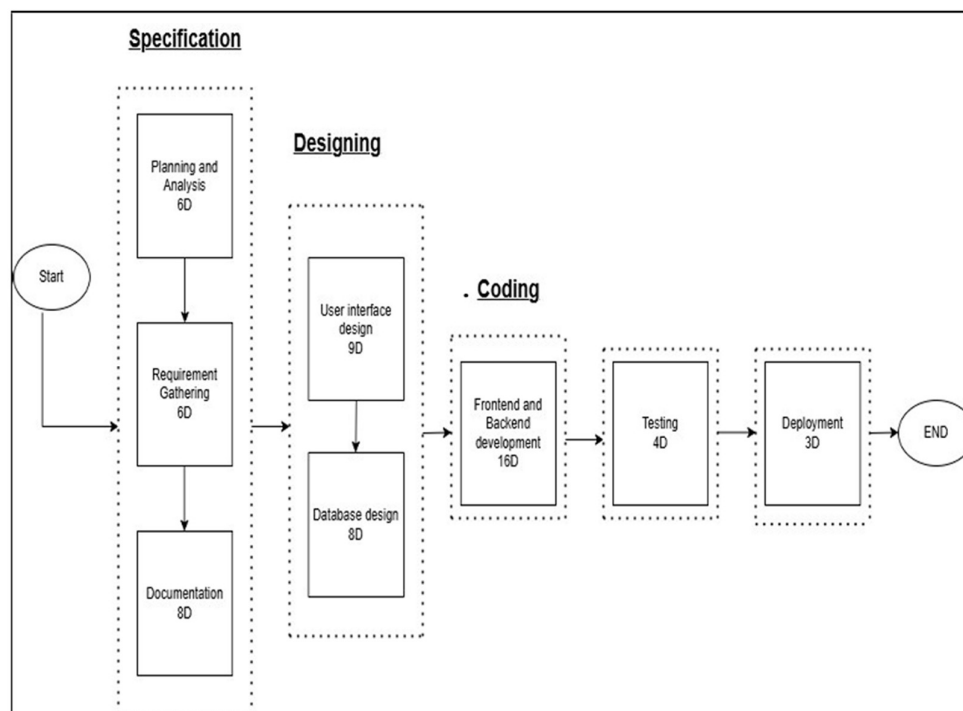
4.8 Methodology

The Software Development Life Cycle (SDLC) model chosen for Career Link was the Incremental Model, selected for its suitability to phased development and gradual feature integration. This model supported continuous improvement and testing while allowing flexibility to build and expand the system step by step. The methodology followed these phases:

- Initial Planning and Requirement Analysis: Identifying foundational requirements such as separate user and company roles, authentication, and company listing.

- **System Architecture Design:** Designing a modular structure to support independent development and integration of each feature (login, add interview info, subscription, search).
- **Incremental Development,** carried out across five increments — (1) basic structure and login/signup, (2) company listing and browse functionality, (3) add interview info and search, (4) subscription and notification integration, and (5) profile pages, UI enhancements, and final polishing.
- **Testing of Each Increment** to ensure proper functionality and integration with existing components.
- **Final Integration and Validation,** where all modules were integrated, system-level testing was performed, and adjustments were made based on user feedback.

An **Activity Chart** supported project planning by visualising task sequences and dependencies and tracking the timeline for each phase—Planning & Analysis, Requirement Gathering, Documentation, UI/UX Design, Database Design, Coding (Frontend and Backend), Testing, and Deployment. A feasibility study covering technical, economic, operational, and legal aspects confirmed the project was practical and achievable using freely available tools and technologies, with minimal financial investment.



Testing followed a combination of Black Box, White Box, and Grey Box techniques, along with Unit Testing of individual modules (login handler, subscription controller, interview CRUD functions), after every feature update, to ensure the platform remained reliable and functionally correct.

4.9 Expected Outcomes

Upon completion, Career Link is expected to deliver a fully functional, responsive web platform where students can register, browse company and interview listings, subscribe to companies for real-time updates, and submit reviews. In contrast, companies can register, manage their profiles, post interview/job listings, and view their subscribers. The platform is expected to reduce the effort required for off-campus job discovery, minimize job-role mismatch through AI-assisted recommendations, and provide both students and companies with a centralized, easy-to-use hiring ecosystem. Cost estimation using the COCOMO model placed the estimated development effort at approximately ₹1,60,000 for a single developer working full-time over a two-month period, reflecting the project's moderate (semi-detached) complexity.

4.10 Certificates of completion and other communication mail proof must be attached

- Offer Letter



- Certificate



5. Bibliography/References

- [1] GeeksforGeeks, "MERN Stack Tutorial," GeeksforGeeks. [Online]. Available: <https://www.geeksforgeeks.org/>.
- [2] JavaTpoint, "Node.js and Express.js Tutorial," JavaTpoint. [Online]. Available: <https://www.javatpoint.com/>.